

ADAM KAEWELL

Github: woodFordR | **Portfolio:** kaewell.com

Github Organization: the-woodford-den | **Email:** adam@kaewell.com | **LinkedIn:** in/adamkaewell

SUMMARY

Senior Backend Engineer with 10+ years designing scalable distributed systems. Specialized in Python microservices, async architectures, and DevOps automation. Proven track record building production systems serving 50,000+ users with comprehensive observability, type-safe APIs, and CI/CD pipelines.

PROFESSIONAL EXPERIENCE

Scala Computing | Senior Backend Engineer

Oct 2024 - Jun 2025

- *Architected Python TUI application (9,673 LOC) using Textual framework for managing Firecracker microVM infrastructure on AWS, enabling real-time terminal access and topology visualization across distributed emulation networks*
- *Designed multi-database SQLite architecture with dynamic per-VM database creation, centralized topology state management (15+ interconnected SQLAlchemy entities), and distributed session tracking with automatic schema migration and connection pooling*
- *Engineered nested SSH tunneling infrastructure enabling terminal access through 3-tier architecture (local → AWS EC2 → Firecracker microVM), implementing deterministic port allocation, automated socat deployment, and hierarchical session cleanup across distributed systems*
- *Developed database-backed pseudo-terminal (PTY) session management system enabling multi-user concurrent terminal access with persistent message queuing, ANSI rendering, and thread-safe async I/O handling*
- *Built multi-stage infrastructure orchestration coordinating Pulumi automation API, AWS EC2 security groups, and custom Platform/Pool Service APIs with stateful progress tracking, automated IP whitelisting, and comprehensive error recovery*
- *Implemented cryptographic SSH key lifecycle management with automated RSA generation (2048-bit), secure permission handling, multi-format key loading (RSA/DSS/ECDSA/Ed25519), and cross-service synchronization*
- *Tech Stack: Python 3.12, Textual, SQLAlchemy, SQLite, Paramiko, asyncio, Pulumi, AWS EC2, Firecracker, Docker, Logfire, Pydantic*

Xometry (Acquired from Maketime 2018)

Software Engineer → Senior Software Engineer

Feb 2015 - Jan 2022

- *CAD Integration & Desktop Development:*
 - *Architected Fusion 360 desktop add-in (Python) enabling instant manufacturing quotes for 50,000+ users, integrating Autodesk's 3D modeling API with Xometry's pricing service through STEP file export, entity token tracking, and bidirectional JavaScript-Python event communication*
- *Frontend Architecture & Engineering Standards:*
 - *Established enterprise React + TypeScript templates (CRA and custom Webpack 5) defining frontend standards across engineering organization, implementing Apollo Client GraphQL integration, Emotion CSS-in-JS, Xometry Loft design system, and FluentJS internationalization infrastructure*
 - *Architected runtime environment variable injection pattern solving dynamic configuration challenges for Kubernetes deployments, enabling environment-agnostic Docker images with shell-based env.js generation at container startup*

- Built custom Webpack 5 configuration with Module Federation for micro-frontend architecture, implementing CDN externalization for React/ReactDOM (~200KB bundle reduction) and content-hash caching strategies
- GraphQL & API Development:
 - Designed type-safe GraphQL development workflow with automated TypeScript code generation from schema, mock server infrastructure for isolated testing, and Puppeteer contract tests validating production builds
- DevOps & Infrastructure:
 - Engineered multi-stage CI/CD pipelines (GitLab + GitHub Actions) with Terraform IaC, Docker builds via Kaniko, Helm deployments to multi-region K8s clusters, and automated S3 artifact distribution with versioned releases
 - Containerized frontend applications with Docker multi-stage builds, Nginx configuration for SPA routing with health endpoints, and cache-control strategies for production optimization
- Testing & Quality:
 - Established multi-layer testing strategy combining Jest + React Testing Library, Puppeteer contract tests, and strict TypeScript/ESLint (Airbnb) enforcement in CI pipelines
 - Tech Stack: Python 3, React 17, TypeScript 4, GraphQL, Apollo Client 3, Webpack 5, Docker, Kubernetes, Helm, Terraform, AWS (S3, ECR), GitLab CI/CD, Nginx, Autodesk Fusion 360 API

PERSONAL PROJECTS

Daredevil | Full-Stack CI/CD Platform

Sept 2025 - Present

GitHub: github.com/the-woodford-den/daredevil-deployer

- Architected async-first FastAPI application with SQLAlchemy ORM, achieving end-to-end type safety from PostgreSQL to React 19 frontend using TypeScript
- Designed production-grade database architecture with async connection pooling (pool_size=10, max_overflow=5), Alembic migrations, and custom DataStore abstraction for transaction management
- Implemented comprehensive GitHub API integration with OAuth2 device flow, RS256 JWT authentication, and webhook handling for automated CI/CD workflows
- Established observability infrastructure using Logfire and OpenTelemetry for distributed tracing across async operations with structured logging
- Built security layer with Argon2 password hashing (OWASP recommended), JWT-based authentication, Redis token blacklisting, and httponly cookie sessions
- Created CI/CD pipeline with GitHub Actions running automated tests, Ruff linting, and database migrations in PostgreSQL service containers
- Tech Stack: Python 3.13, FastAPI, PostgreSQL 16, Redis, React 19, TypeScript, SQLAlchemy, Alembic, pytest, GitHub Actions, Logfire, OpenTelemetry